10.4 Inscribed Angles and Polygons

Name _____ Period ____ Date _____

Geometry 10.4 · Student sheet

I can…

- find the measure of an inscribed angle or its intercepted arc.
- use the fact that a triangle inscribed in a semicircle is a right triangle.
- use the fact that opposite angles of an inscribed quadrilateral are supplementary.

Vocabulary · fill in the missing word

Inscribed angle — an angle whose vertex is on a circle and whose sides _____ chords of the circle

Intercepted arc — the arc that lies in the interior of an _____ angle and has endpoints on the angle

Inscribed polygon — a polygon whose _____ all lie on a circle

Circumscribed circle — the circle that contains all the vertices of an _____ polygon

Part 1 · Discovery: angles on the circle

Draw a circle with center O. Mark A and B on it and draw ∠AOB. On the other side of the circle, mark C, D, and E, and draw ∠ACB, ∠ADB, and ∠AEB.

∠AOB	∠ACB	∠ADB	∠AEB

What do you notice?

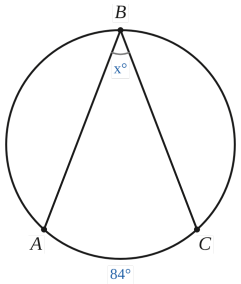
Part 2 · Notes and examples (fill in each note, then work the example)

Measure of an Inscribed Angle Theorem: the measure of an inscribed angle is _____ the measure of its intercepted arc.

Example 1

a. Find m∠ABC.

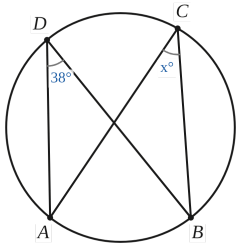
b. An inscribed angle measures 55°. What is the measure of its intercepted arc?



Inscribed Angles of a Circle Theorem: if two inscribed angles of a circle intercept the same arc, then the angles are _____.

Example 2

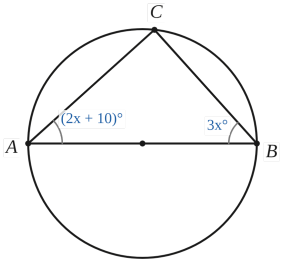
m∠ADB = 38°. Find m∠ACB and m(arc AB).



Inscribed Right Triangle Theorem: a triangle inscribed in a circle is a right triangle if and only if one side is a _____.

Example 3

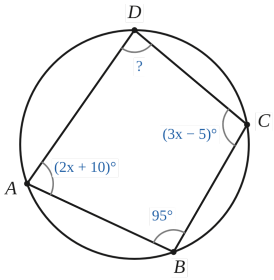
$\overline{AB}$ is a diameter of the circle. Find the value of x. Then find m∠A and m∠B.



Inscribed Quadrilateral Theorem: a quadrilateral can be inscribed in a circle if and only if its opposite angles are _____.

Example 4

Quadrilateral ABCD is inscribed in the circle. Find the value of x. Then find m∠D.



Example 5

Can a rhombus that is not a square be inscribed in a circle? Explain.

Part 3 · Partner practice

Partner 1

An inscribed angle intercepts an arc of 130°. Find the measure of the angle.

Partner 2

Quadrilateral PQRS is inscribed in a circle. m∠Q = (4x + 8)° and m∠S = (2x + 10)°. Find x, m∠Q, and m∠S.